

Fact Verification in Large Language Models

Fact verification checks whether statements generated by a language model are supported by reliable evidence. This document focuses on the verification workflow, not on the causes of model errors.

A practical verifier first splits an answer into factual claims. Each claim should be small enough to compare with a passage. For example, a sentence containing a model name and a year should be checked as a separate claim from a sentence describing an application.

The next step is evidence retrieval. A retriever should return passages that contain the exact information needed for the claim. A passage that is only topically related should not be treated as enough support for a name, date, number, benchmark, founder, method, or definition.

Support scoring assigns labels such as supported, weak_support, and unsupported. Supported means the evidence directly states the claim or clearly entails it. Weak support means the evidence is related but incomplete. Unsupported means the evidence does not contain the fact, or the claim conflicts with the evidence.

Natural language inference can be added as an optional verification layer. An NLI model compares an evidence passage with a claim and predicts entailment, contradiction, or neutral. NLI is useful for entity, date, and number disagreements, but it can be expensive on low-memory machines.

A good verification report should show the claim, the best evidence passage, the support label, important rule flags, and the final confidence. This makes the result explainable for research evaluation and for user-facing debugging.